Question
When starting a project in a technology stack you have never used before, what is your typical first step?

Options

A. Read official documentation and architecture guides first
B. Build a small prototype to experiment with the technology
C. Watch tutorials or study existing open-source projects
D. Ask experienced teammates for recommended best practices

Trait Signals

A → LearningAgility +0.8, ResearchEfficiency +0.9
B → LearningAgility +0.7, AnalyticalDecisionMaking +0.5
C → LearningAgility +0.6
D → CollaborationStyle +0.7

2. Handling a production bug

Question
A critical production bug begins affecting users. What is your first action?

Options

A. Roll back the latest deployment to restore service
B. Analyze logs and traces to identify the root cause
C. Deploy a quick patch to reduce user impact immediately
D. Attempt to reproduce the issue in a staging environment

Traits

A → IncidentManagement +0.7
B → DebuggingDepth +0.9
C → SystemSustainability +0.6
D → DebuggingDepth +0.7

3. Managing technical debt

Question
How should technical debt be handled in a fast-paced development team?

Options

A. Focus on shipping features and address debt only when necessary
B. Allocate time in every sprint specifically for refactoring

C. Refactor code whenever you modify an existing component
D. Plan periodic engineering cleanup cycles

Traits

A → TechnicalDebtStrategy +0.4
B → SystemSustainability +0.8
C → TechnicalDebtStrategy +0.7
D → SystemSustainability +0.7

4. Applying for a role with an unfamiliar tool

Question
A job description lists a tool you have never used before. How do you approach the opportunity?

Options

A. Explain transferable skills and willingness to learn quickly
B. Build a small project using the tool before the interview
C. Study the tool's ecosystem and architecture beforehand
D. Avoid applying until you gain formal experience

Traits

A → SelfAwareness +0.7
B → Adaptability +0.8
C → ResearchEfficiency +0.7
D → SelfAwareness +0.4

5. Priorities during code review

Question
When reviewing a teammate's code, what do you look for first?

Options

A. Logical correctness and potential bugs
B. Code readability and maintainability
C. Performance bottlenecks or inefficiencies
D. Security vulnerabilities

Traits

A → CodeQualityFocus +0.8
B → SystemSustainability +0.8

C → TradeoffReasoning +0.7
D → SecurityAwareness +0.9

6. Feature impacting performance

Question
A client requests a feature that will slow the system by about 40%. What would you do?

Options

A. Implement it because it is technically feasible
B. Explain the performance trade-off and propose alternatives
C. Implement it while optimizing other parts of the system
D. Escalate the decision to product leadership

Traits

A → StakeholderCommunication +0.3
B → TradeoffReasoning +0.9
C → ArchitectureThinking +0.6
D → StakeholderCommunication +0.6

7. Architecture preference

Question
Which system architecture do you generally find most robust for scalable applications?

Options

A. Modular monolith architecture
B. Microservices architecture
C. Serverless architecture
D. Event-driven distributed architecture

Traits

A → ArchitectureThinking +0.5
B → ArchitectureThinking +0.8
C → ArchitectureThinking +0.6
D → ArchitectureThinking +0.9

8. Unrealistic deadlines

Question
You realize halfway through a project that the deadline cannot realistically be met.

Options

A. Inform stakeholders immediately and propose revised estimates
B. Work extra hours to try meeting the deadline
C. Reduce feature scope quietly to finish on time
D. Wait until closer to the deadline before raising concerns

Traits

A → Accountability +0.9
B → Ownership +0.6
C → CommunicationTransparency +0.2
D → Accountability +0.3

9. Ideal role balance

Question
In your ideal engineering role, how would your time be distributed?

Options

A. Mostly coding and implementing features
B. Balanced between coding and architecture decisions
C. Mostly system design and architecture
D. Leading technical strategy and mentoring engineers

Traits

A → CareerOrientation +0.4
B → CareerOrientation +0.6
C → LeadershipMaturity +0.6
D → LeadershipMaturity +0.9

10. Security vulnerability discovery

Question
You discover a security vulnerability in a module unrelated to your current task.

Options

A. Fix the vulnerability immediately
B. Report the issue and create a ticket

C. Inform the team and suggest potential solutions
D. Ignore it because it is outside your task scope

Traits

A → Ownership +0.9
B → SecurityResponsibility +0.7
C → Ownership +0.7
D → Ownership +0.1

11. Privacy regulation pressure

Question
Your team leader suggests ignoring a minor privacy regulation during testing.

Options

A. Follow instructions because it is only for testing
B. Express concern and recommend compliance
C. Escalate the issue to appropriate leadership
D. Refuse to proceed until the issue is resolved

Traits

A → Integrity +0.1
B → ComplianceAwareness +0.8
C → Integrity +0.9
D → Integrity +1.0

12. Technical disagreement

Question
You disagree with a teammate about a technical design decision.

Options

A. Present benchmarks or data supporting your argument
B. Ask a senior engineer to mediate
C. Build small prototypes to compare both approaches
D. Accept the teammate's decision to avoid conflict

Traits

A → AnalyticalDecisionMaking +0.8
B → CollaborationStyle +0.6
C → AnalyticalDecisionMaking +0.9
D → CollaborationStyle +0.4

13. Accessibility in UI design

Question
When designing a user interface, how do you approach accessibility standards?

Options

A. Only address accessibility when required
B. Follow accessibility guidelines during development
C. Include accessibility checks in testing and QA
D. Prioritize inclusive design from the beginning

Traits

A → UXEmpathy +0.2
B → InclusiveDesign +0.7
C → InclusiveDesign +0.8
D → InclusiveDesign +1.0

14. Handling project failure

Question
A project you led fails after deployment. What is your primary focus in the post-mortem?

Options

A. Identifying technical causes of failure
B. Improving engineering processes and workflows
C. Understanding team decision-making errors
D. Restoring system stability and reliability

Traits

A → ProcessMaturity +0.6
B → ProcessMaturity +0.9
C → LeadershipMaturity +0.6
D → Resilience +0.7

15. Senior vs junior engineers

Question
What primarily differentiates a senior engineer from a junior engineer?

Options

A. Experience with more technologies
B. Ability to design scalable systems
C. Ownership of complex problems
D. Mentoring others and guiding technical decisions

Traits

A → LeadershipMaturity +0.3
B → ArchitectureThinking +0.7
C → Ownership +0.8
D → LeadershipMaturity +0.9

Trait Calculation

```
from collections import defaultdict


def compute_trait_scores(answers, mapping):


    trait_scores = defaultdict(float)

    trait_counts = defaultdict(int)


    for qid, option in answers.items():


        traits = mapping[qid][option]


        for trait, weight in traits.items():

            trait_scores[trait] += weight

            trait_counts[trait] += 1
```

```
normalized = {}

for trait in trait_scores:
    normalized[trait] = trait_scores[trait] / trait_counts[trait]

return normalized
```